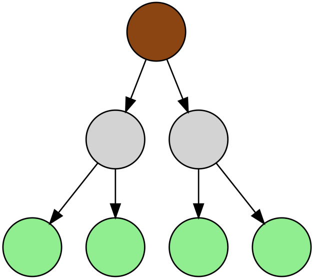
Decision Trees and Random Forest

Comp. Bio II

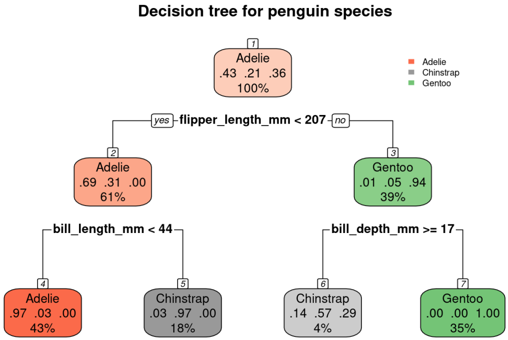
5th May, 2026

Trees



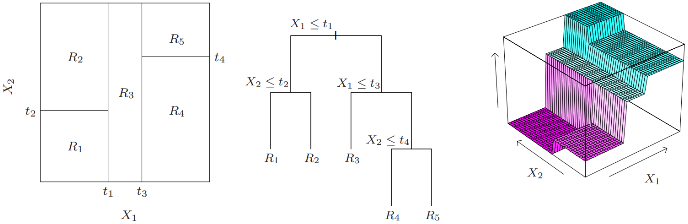
- A **tree** is a connected graph with no cycles.
- A **rooted tree** has a special node called the **root**.
- The root has no parent. All other nodes have exactly one parent.
- **Leaves** are nodes with no children.

Decision tree



- Internal nodes split the data based on feature values.
- Leaves assign a constant prediction (e.g. class label).

Decision tree - visualization



Hastie et al.: *The Elements of Statistical Learning*, 2009

- A decision tree models a piecewise constant function.
- Can be used for both regression and classification.

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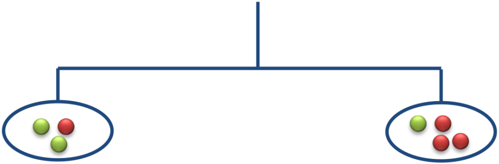
- Stop when a stopping criterion is reached (e.g. small nodes).
- Many variations exist (e.g. limiting depth, random feature selection).

Gini index



- Gini index is defined as

$$G = 1 - \sum_{k=1}^{K} p_k^2$$

where p_k is the fraction of class k among samples in the node.

- Equals the probability of misclassification (if labels are assigned according to class proportions).
- Used to train classification trees.

Bias-variance trade-off

- **Shallow trees**: low variance, high bias (may miss complex patterns).
- **Deep trees**: low bias, high variance (prone to overfitting).
- The trade-off can be controlled by model parameters (e.g. tree depth, minimum node size).

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- Can we get the best of both worlds?

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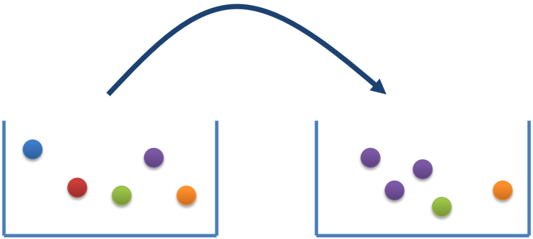
- **Boosting**:
  - Sequentially train *low variance, high bias* models (shallow trees).
  - Subsequent models learn to fix the mistakes of the previous ones.
  - Exploits dependence between models.

Bootstrapping



(Breiman 1996, *Machine Learning*)

- Resampling from the original training set (with replacement).
  - Same number of observations.
  - Some observations may be sampled multiple times.
  - Some observations may not be sampled at all.
- Each resampling produces a slightly different version of the original data.

Random Forest

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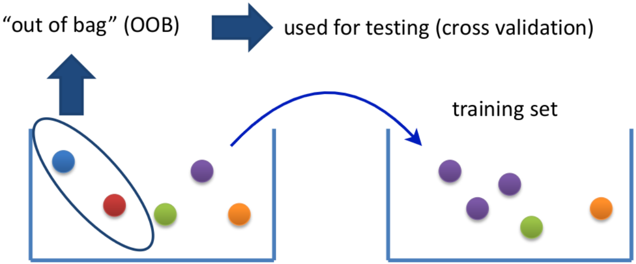
- Additional randomness: at each split, consider only a random subset of features.
  - Reduces correlation between trees.
  - May also prevent trees from getting stuck in local optima during training.

Out-of-bag (OOB) samples



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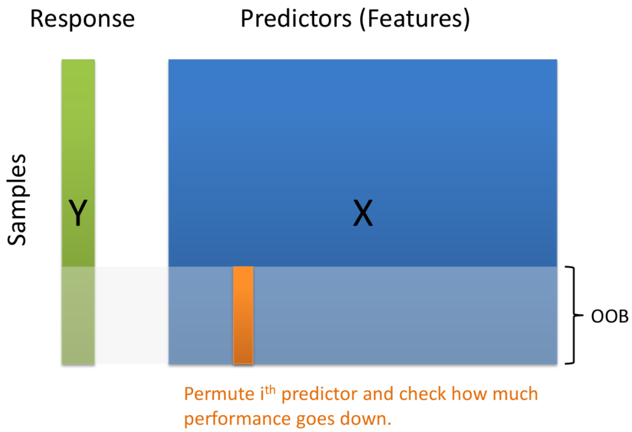
- For every tree, observations not sampled by bootstrap (out-of-bag, OOB) may be used as a test set.
- For each observation, aggregate predictions from the trees where it was out-of-bag.
    - Provides an estimate of the test error of the whole random forest.

Feature importance



- Randomly permute a feature in OOB samples (breaking its relationship with the outcome) and measure the decrease in prediction performance.

Boosting

- **AdaBoost** (adaptive boosting):
  - Sequentially train very shallow trees (so-called *decision stumps*).
  - Models aim to minimize a weighted loss (samples have weights).
  - Misclassified samples get higher weight, so future models focus on them.
  - Combine models using a weighted average (better models get higher weight).

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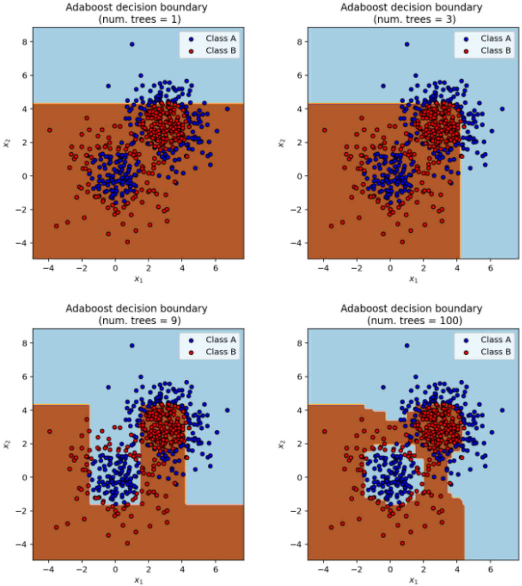
- **Gradient boosting** (e.g. XGBoost):
  - Similar idea: sequentially add models to correct previous errors.
  - Fit new models to the gradients of the loss (for RSS, this corresponds to residuals).
  - Uses small trees (deeper than decision stumps).

AdaBoost visualization



The End

Acknowledgment:

- Adapted from Prof. Andreas Beyer's slides.
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- Tree drawings are AI-generated (ChatGPT).